



Alpha-8 Staircase Tower

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IS 3696

IS 4014

IS 1161

ISO 9001

Cuplock scaffolding engineered for self-draining performance.

The **Alpha Staircase Tower AST-8** is engineered for sites where drainage is critical — basement excavations, below-grade construction, open-air infrastructure, and wet or muddy environments. The AST-8 configuration (8-leg cuplock) features M.S. Square Pipe Grating treads and M.S. Rectangular Pipe Grating landing platforms that eliminate mud and water accumulation for safer access in adverse conditions.

The 60 OD mm welded socket joint provides positive vertical continuity without loose spigot pins, accelerating erection and improving structural integrity. Guardrails are provided where applicable to enhance user safety, while the uppermost platform is equipped with full fall protection. Adjustable base jacks and mandatory tie bars ensure secure installation on varied terrain.

PIPE GRADE

IS 3502 for square pipe grating

TUBE OD

48.3 mm M.S.

PAINT FINISH

Red-Oxide + Blue Enamel

STAIR SURFACE

M.S. Square Pipe Grating

GRID SIZE

1,800 x 3,800



AST-8 • 8-Leg Cuplock Staircase Tower in service

Overall Tower Geometry

GRID SIZE (L × W) 3,800 × 1,800 mm	PLATFORM HEIGHT Up to 60,000 mm (60.0 m)
TOWER OVERALL HEIGHT Platform height + 1,500 mm (top rail)	NO. OF VERTICAL LEGS 8 legs

Stair Geometry

Parameter	Specification
Stair Width (clear)	700 mm
Railing both sides	Yes
Stair Material	M.S. Square Pipe Grating — 25 × 25 mm (open mesh, self-draining)
Side Railing Height (Stair Flights)	940 mm (both sides) — M.S. Square Pipe
Guard Rail / Mid Rail (Stair Flights)	470 mm (both sides) — M.S. Square Pipe
Top Platform: Side Railing Height	1,500 mm (both sides)
Top Platform: Guard Rail (Mid Rail)	1000 mm (both sides)
Step Rise	206 mm (Class-B per EN 12811-1; BS 1139-3)
Tread (going)	270 mm (Class-B per EN 12811-1; BS 1139-3)
Staircase Angle	41° (EN 12811-1: 30–55°; OSHA 1926.451: 40–60°)
Step Arm UDL (Design Load)	450 kg/m² (2.45 kN/m² 50.1 psf)
No. of stairs per flight	7

Landing Platform — Quick Reference

LANDING WIDTH 1,800 mm	LANDING DEPTH 1,000 mm	TOE GUARDS Not provided — economy
LANDING FIXING Hook-on, secure	PLATFORM UDL 250 kg/m²	SURFACE M.S. Rectangular Pipe Grating

Structural Framing

Component	Specification
Cuplock Vertical (Standard)	48.3 OD × M.S. IS 1161 Grade YST210
Top Cup	SG / Malleable Cast Iron
Bottom Cup	Pressed Metal
Cup Spacing	500 mm centers
Cuplock Ledger	48.3 OD M.S. pipe, forged ledger blades each
Vertical Joint (Socket)	60 mm OD socket, welded onto vertical standard
Bracing Pipe	48.3 OD M.S. pipe, all 4 sides
Couplers & Clamps	Heavy-duty forged; IS 2750 / EN-74 compliant

Base & Anchorage

Component	Specification
Adjustable Base Jack	Spindle 38 mm OD; Total 600 mm; Adjustable 450 mm; Base Plate 150×150×6 mm M.S.
Tie Bar	48 mm OD M.S. pipe + welded holding plate 150 x 150 x 6 mm ; anchor fastening provision; secures

Applicable Standards & Design Codes

Standard	Scope / Relevance
IS 1161:2014	Steel tubes for structural purposes — OD, wall thickness & mechanical properties (YST210) for all M.S. tubes.
IS 2750:1964	Steel scaffolding — fittings. Drop-forged and pressed couplers and clamps.
IS 3696 (Pt.1):1987	Safety code for scaffolds & ladders — stairway design, railings, platform protection.
IS 4014 (Pt.1):1967	Steel tubular scaffolding — materials. Component dimensions, grades, workmanship.
IS 4014 (Pt.2):1967	Steel tubular scaffolding — safety regs. Stairway access, tie-bar spacing, bracing.
EN-74	Couplers for steel scaffolding — strength and dimensional requirements for drop-forged couplers.
ISO 9001:2015	Quality management systems — Prime Steeltech QMS certification.

fall

Key Features & Applications

SAFETY NOTICE — TOE BOARDS NOT PROVIDED (ECONOMY CONFIGURATION)

Toe boards on intermediate landings are NOT provided in this economy configuration. The contractor and site engineer must ensure supplementary measures to prevent tools and materials falling from landing edges are in place before permitting personnel access. A project-specific risk assessment by a competent person is mandatory.

Cuplock System

Tool-free assembly — cup locks up to 4 ledgers per node simultaneously.

Square Pipe Grating Treads

M.S. 25 × 25 mm open-mesh; self-draining, eliminates mud/water on treads.

Rectangular Grating Platforms

M.S. 25 mm open-mesh landings; lightweight, easy site handling.

Welded 60 mm Socket Joint

Positive vertical continuity; faster erection, no loose spigot pins.

Self-Draining Construction

Open grating ideal for basement, excavation & exposed wet-site use.

Platform UDL

250 kg/m² design load (2.45 kN/m² | 50.1 psf); personnel & tools access.

Adjustable Base Jack

150×150×5 mm plate; 450 mm adjustable travel; levels uneven ground.

Tie Bar to RCC / Walls

48 mm OD tie bar with anchor fastening; mandatory beyond 6.0 m height.

Typical Applications

- Basement excavation & below-grade construction
- Infrastructure: bridges, flyovers, metro viaducts
- Open-air & exposed infrastructure
- Demolition & decommissioning works
- High-rise residential & commercial construction
- Industrial plants: refineries, petrochemical, power
- Oil & gas platforms, jetty structures
- Short-duration construction access

Safety Guidelines

SAFETY NOTICE

Never exceed the Safe Working Load (SWL) for the grating and span combination. Guardrails and mid-rails must be installed before access at height. Scaffold erection, alteration and dismantling by trained personnel only (IS 3696). Conduct a project-specific risk assessment before permitting personnel access.

- Inspect every grating panel before installation — remove cracked, bent or heavily corroded units immediately.
- Ensure hook-on fittings are fully seated on the tower structure before personnel access.
- Keep tread and landing grating free of mud, debris, or obstructions at all times, even though the mesh is self-draining.
- Do not mix component types or sizes within the same bay without written engineering approval.
- Tie bars must be fixed to the RCC or structural frame at intervals per structural design; mandatory once height exceeds 6.0 m.
- Display inspection tags (load class, date, inspector name) at each access point.
- Field welding of repairs is NOT permitted — return damaged components for factory repair.
- Toe boards are NOT provided on intermediate landings in this economy configuration — arrange supplementary fall-object protection before use.

Notes & Disclaimers

- Staircase towers must be designed, erected, and inspected by competent persons in accordance with IS 3696 Part 1 and IS 4014 Parts 1 & 2,
- Toe boards on intermediate landings are NOT provided in this economy configuration. Supplementary measures to prevent falling objects must be in place before permitting access.
- Tie bars are mandatory for heights exceeding 6.0 m; install at maximum 3.0 m vertical and horizontal spacing. Spacing must be confirmed by a structural engineer.
- Adjustable base jack spindle: 38 mm OD pipe; total length 600 mm; usable/adjustable length 450 mm. Must not be extended beyond usable length; use sole boards on soft or unpaved ground.
- M.S. grating treads and platforms must be inspected regularly for corrosion, deformation, weld cracking, or mechanical damage. Damaged panels must be replaced before further access.
- Platform and step-arm UDL of 250 kg/m² (2.45 kN/m²) is the maximum design value. Site loadings must not exceed this value.
- This datasheet is for information and quotation reference only. Structural calculations and certified drawings are prepared separately. Contact Prime Steeltech for project-specific design.

Contact & Further Information

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